Project Proposal: Data Management (2025/2026)

Group Members

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Type of Project

[NoSQL] Select a dataset, a graph database tool, and a relational DBMS, and compare the results of analyses on differentsystems, highlighting advantages and disadvantages of each approach in terms of efficiency
Focus: ability to compare different technologies

Description of the Dataset

I intend to use the Social Network: Reddit Hyperlink Network dataset provided by the Stanford Network Analysis Project (SNAP).

- **Dataset Reference Link:** https://snap.stanford.edu/data/soc-RedditHyperlinks.html
- **Overview:** The Reddit Hyperlink Network dataset from the Stanford Network Analysis Project (SNAP) models directed inter-community interactions across Reddit over a 2.5-year timespan (January 2014 – April 2017), mapping 55,863 unique subreddits as nodes and 858,490 post-to-post hyperlinks as edges. Distributed across distinct files for post body links, post title links, and 51,278 subreddit embeddings, the network is formally classified as directed, signed, temporal, and attributed. Every hyperlink traces a distinct connection from a source to a target community, carries a precise chronological timestamp, includes a multi-dimensional text property vector capturing the linguistic features of the source post, and is annotated with a discrete edge weight of -1 (explicitly hostile/negative) or +1 (neutral/positive) derived from a high-accuracy, crowdsourced classifier.

Brief Description of the Work

The goal of this project is to evaluate and compare the structural efficiency, scalability, and query language ergonomics of a Relational DBMS (PostgreSQL) against a native Graph Database (Neo4j) using the Reddit Hyperlink Network. To avoid trivial modeling, we will design a multi-entity relational schema (using normalized tables and foreign keys) alongside an equivalent multi-label graph schema in Neo4j representing subreddits, posts, and distinct interaction types. I will benchmark execution times and system resource consumption across analytical queries of increasing complexity: simple point lookups, 2-hop collaborative filtering joins, and complex recursive path-finding (such as cycle detection to identify closed "echo chambers" or mutual toxic harassment loops between subreddits). Finally, I will conduct a qualitative analysis contrasting SQL recursive Common Table Expressions against native Cypher traversal syntax, evaluating expressive power, code readability, and developer ease-of-use.